

Aryan Ahlawat

Bachelor of Computing (Honours) • AI Specialization • Economics Minor

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001 EDUCATION

Queen's University Bachelor of Computing (Honours), AI Specialization Stream

Kingston, ON • 2024 – Present

002 EXPERIENCE

Software Developer Co-op [Co-operators](#)

May 2026 – Aug 2026

AWS Lambda • Lex • Python • Postman

- Developing **RESTful Prefill API integrations** to enable real-time auto-population of insurance quoting forms, orchestrating data retrieval across multiple internal services and third-party endpoints to eliminate manual entry and reduce quoting workflow completion time
- Driving spec driven development for APIs, building automation that generates reference docs directly from **OpenAPI specifications** so published documentation stays in sync with service contracts without manual upkeep

Design Team Engineer [QMIND](#)

Sep 2025 – Apr 2026

LangChain • HuggingFace • RAGAS • Python

- Architected the **vector storage** and **conversational memory** infrastructure for an agentic **RAG** pipeline, engineering an **episodic memory** layer that deduplicates semantically relevant queries, while persisting context across multiple queries
- Engineered a hybrid retrieval pipeline using **Pinecone**, merging **BM25 sparse search** and **dense semantic embeddings** with **reciprocal rank fusion (RRF)** to maximize context recall
- Implemented a defensive **multi-hop failsafe** mechanism that evaluates context sufficiency, automatically decomposing complex queries and executing secondary retrieval hops when initial evidence is weak

Machine Learning Co-op [Acetech](#)

May 2025 – Aug 2025

PyTorch • Pandas • SQL • NumPy

- Developed **Multi-Layer Perceptron (MLP)** regression models in PyTorch to predict lab test durations, tuning hyperparameters to achieve a **12% reduction** in Mean Squared Error loss against the previous baseline
- Preprocessed and engineered features from an extensive SQL dataset; built automated data cleaning pipelines with **Pandas, SQLAlchemy, and NumPy** to impute missing values and detect outliers

Drive Control Developer [Queen's Knights Robotics](#)

Oct 2024 – Mar 2025

FreeRTOS • MATLAB • C

- Programmed low-level robotic drive-control systems in C, implementing **PID controllers** to ensure smooth acceleration and precise velocity control
- Developed motion algorithms that improved autonomous navigation efficiency, reducing path deviations for competition tasks by almost **20%**

003 PROJECTS

CardIQ [Hack the 6ix](#) [devpost ↗](#) [github ↗](#)

Jul 2026

Python • FastAPI • PuLP/CBC • Pydantic • StreamLit

- Built a deterministic multi-card payment router, formulating spending allocation as an **integer program (PuLP/CBC)** that optimizes for hard credit limits, utilization ceilings, and sign up bonus constraints, with a greedy solver as a verified fallback
- Fine-tuned a **small language model** on synthetic data to translate plain-text financial goals into the exact mathematical weights required by the solver, enabling natural language routing and verifying decision accuracy

Visualizelt [QHacks](#) **WINNER • MAYOR'S INNOVATION AWARD** [devpost ↗](#)

Feb 2025

YOLOv8 • PyTorch • Stable Diffusion • OpenCV

- Engineered a real-time augmented reality pipeline using **YOLOv8 Nano** and **Mask R-CNN** for rapid object detection, integrating **MediaPipe** and **OpenCV ORB feature matching** for continuous, low-latency tracking
- Integrated **Stable Diffusion** to generate custom textures, replacing manual editing by dynamically perspective-warping and alpha-blending stylized AR graphics directly onto moving targets in live video feeds

004 SKILLS & CERTIFICATIONS

Languages • Frameworks Python, Java, C, SQL, JavaScript, PyTorch, LangChain, FastAPI, Flutter, Scikit-Learn, OpenCV

Tools • Platforms AWS, Azure, Oracle Cloud, Git, Postman, OpenAPI, MATLAB, Databricks, Docker

Certifications Oracle Cloud Infrastructure Foundations, Microsoft AZ-900, NVIDIA Building RAG Agents with LLMs